TUDLO — Product Requirements Document (PRD)

Hackathon Edition | SDG 4: Quality Education | Philippines

I. UNDERSTAND THE PROBLEM

A. Read the Challenge

Hackathon Theme: SDG 4 — Quality Education in the Philippines

Judging Criteria Alignment:

Criterion	Weight	Tudlo's Angle
Impact & Sustainability	40%	Addresses recurring, permanent disaster-related learning continuity gaps

MVP Quality & Execution	35%	Scoped, offline-first, teacher-centered curriculum recovery tool
Technical Complexity	20%	Modular PWA, disruption-aware logic, optional weather-trigger integration, aggregated recovery data layer
Team Engagement	5%	Documentation, guidelines, submission requirements

B. Define the Problem

What problem are you solving?
When classes are disrupted by typhoons or other emergencies, Philippine public school teachers lose track of exactly where each class stopped in the curriculum. When school resumes, teachers often have to guess where to continue, restart lessons already covered, or skip ahead. This wastes instructional time, causes learning gaps, and makes recovery inconsistent across classrooms and schools.

Who experiences this problem?

- **Primary:** Public school teachers in disruption-prone regions
- **Secondary:** Students in those classrooms who experience inconsistent recovery and learning gaps
- **Tertiary:** School principals, division superintendents, and DepEd offices that lack structured visibility into curriculum disruption and recovery

Why does this problem matter?
The Philippines experiences recurring school disruptions due to typhoons and other emergencies. Even a few missed days per event can compound into a substantial loss of instructional continuity across a school year. Without a structured recovery system, those missed days are often handled through memory, informal messaging, or manual notes, which are not reliable across repeated disruptions.

How is it solved today?
It is not solved consistently. Teachers rely on personal memory, informal Messenger group

chats asking “Saan na tayo?”, handwritten notes, or spreadsheets that are not disaster-hardened and are not aggregated for school- or division-level visibility.

Existing Tool	What It Does	What It Cannot Do
LIS	Tracks enrolled learners	Cannot track curriculum position
DepEd Commons	Hosts curriculum content	Cannot track where a class is in that content
BEIS / EBEIS	Annual school statistics	No real-time disruption tracking
SDRRM Forms	Documents physical damage	Does not capture instructional continuity loss
Google Classroom	Assignment management	Requires internet and does not manage disruption recovery
SLMs (Printed Modules)	Content during closure	No resumption coordination after reopening
Messenger / Facebook	Teacher communication	No structured data or curriculum logic
Excel / Google Sheets	Informal tracking	Not disaster-hardened and not aggregated

C. Output

Problem Statement

Filipino public school teachers in disruption-prone areas have no persistent, disaster-hardened record of their class's exact curriculum position. Each closure resets operational awareness, forcing teachers to guess where to resume, which wastes instructional time and creates cumulative learning gaps.

Target Users

- **Primary:** Public school teachers in disruption-prone regions
- **Secondary:** School principals and division superintendents
- **Tertiary:** DepEd regional offices and national curriculum planners

Success Metrics

Metric	Target
Time to resume the correct lesson after a disruption	Under 60 seconds
Reduction in redundant re-teaching	At least 3 instructional days recovered per disruption event
Teacher adoption rate in pilot schools	70% active use within the first quarter
Disruption events logged and visible to admins	Within 24 hours of the event
Recovery visibility for support planning	Schools with high disruption density can be identified quickly for intervention

II. IDENTIFY THE SUSTAINABLE DEVELOPMENT GOALS (SDGs)

A. Select Relevant SDGs

Primary SDG:
SDG 4 — Quality Education
"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all."

Supporting SDGs:

- **SDG 13 — Climate Action**: Tudlo addresses the educational consequences of recurring climate disruption.
- **SDG 10 — Reduced Inequalities**: Disaster-prone and lower-income communities are disproportionately affected by learning interruptions.

B. Define the Project's Contribution

SDG 4 Target	How Tudlo Contributes
4.1 — Free, equitable, quality primary and secondary education	Reduces curriculum coverage inequality between disrupted and non-disrupted classrooms
4.5 — Eliminate education disparities for vulnerable populations	Supports disaster-prone communities that are consistently left behind
4.c — Increase supply of qualified teachers and support teacher quality	Gives teachers a tool that helps them recover instruction and manage continuity

Measurable Impact:

- Recovery of instructional time after each disruption event
- Better visibility into which classes need support

- Aggregated recovery patterns that can inform long-term educational continuity planning

C. Align with the Hackathon Theme

Tudlo is designed for a reality where disruption is recurring, not exceptional. The project is not only about typhoons; it is about any interruption that prevents teaching continuity. Typhoons are the primary MVP case because they can be automatically detected or inferred through weather-triggered logic, making the system easier to demonstrate and validate. The broader design remains open to floods, earthquakes, fires, evacuations, transport disruptions, and other causes of class interruption.

Selected SDGs: SDG 4 (Primary), SDG 13, SDG 10 (Supporting)

SDG Alignment Statement:
Tudlo advances SDG 4 by ensuring that repeated school disruptions do not permanently erase a teacher's knowledge of where a class stands. By supporting precise resumption and catch-up planning, the system transforms disruption from an educational setback into a recoverable event.

Expected Social Impact:

- Immediate: Teachers resume correctly within minutes of reopening
- Medium-term: School leaders gain visibility into recovery status and can identify classes needing support
- Long-term: Aggregated data supports evidence-based educational continuity planning

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III. RESEARCH

A. Study Existing Solutions

Tool / Solution	Type	Strength	Critical Gap
DepEd Commons	Government content platform	Comprehensive curriculum content	No position tracking and no disruption feature

LIS	Government learner database	Complete learner records	No curriculum awareness
BEIS / EBEIS	Government school database	School-level statistics	Annual updates only; not real-time
SDRRM Forms	Government disaster reporting	Documents physical damage	Does not capture instructional loss
Google Classroom	Private LMS	Assignment management	Requires internet and is not disruption-aware
Self-Learning Modules (SLMs)	Government print materials	Offline content delivery	No recovery coordination after reopening
Messenger / Viber	Consumer messaging	Easy adoption	No structure, no data, no curriculum logic
Khan Academy / Coursera / EdX	Global EdTech	Content delivery	Not aligned to Philippine K–12 public school disruption recovery

B. Gather Insights

Pain Points:

- Teachers rely on memory to resume after disruptions
- There is no official tool that tracks curriculum position in real time

- Post-disruption stress makes complex tools unrealistic
- Rural areas often have low or intermittent internet connectivity
- DepEd lacks a structured signal for which schools need support after disruptions

User Needs:

- A tool useful on ordinary school days, not only during crises
- Zero-internet capability
- Minimal daily time investment
- Clear guidance on what to do next, not just raw data
- Easy installation without app-store dependence

Market Gap:
No existing tool tracks curriculum position at the classroom level in a disaster-hardened, offline-capable, recovery-ready format.

Technical Constraints:

- Low-end Android devices
- Intermittent or absent mobile data
- Limited device storage
- Variable digital literacy among teachers
- No guaranteed access to app stores

C. Output

Competitive Analysis: See table above

Opportunity List:

1. Curriculum position tracking as a new data primitive for Philippine education
2. Offline-first PWA as the distribution and access model
3. Typhoon-triggered disruption detection as an MVP automation layer
4. Recovery graph and readiness view for schools with repeated disruptions
5. Aggregated recovery data for intervention planning
6. Long-term roadmap that dynamically spaces lessons using curriculum references and disruption history

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IV. BRAINSTORM SOLUTIONS

A. Ideas Generated

1. SMS-based lesson check-in system for zero-internet environments
2. Printed curriculum passport booklet updated by teachers weekly
3. Offline PWA curriculum tracker with cloud sync
4. Integration layer on top of DepEd Commons adding position tracking
5. Messenger bot that accepts lesson log commands
6. QR-code-based lesson completion stamps on SLM modules
7. Radio-broadcast lesson resumption announcements after major disruptions
8. Excel template distributed through DepEd division offices

B. Evaluate

Idea	Impact	Feasibility	Innovation	Demo Potential
SMS-based system	High	Medium	Medium	Low
Printed curriculum passport	Medium	High	Low	Low
Offline PWA tracker	High	High	High	High
DepEd Commons integration layer	High	Low	Medium	Medium
Messenger bot	Medium	Medium	Medium	Medium

QR-code SLM stamps	Medium	Medium	Medium	Low
Radio announcements	Low	Low	Low	Low
Excel template	Low	High	Low	Low

C. Selected Solution

Tudlo — An offline-first Progressive Web App (PWA) that lets teachers record daily curriculum progress, freeze classroom state during any disruption, generate exact resumption instructions, and produce catch-up plans when classes reopen. Typhoons are the primary MVP trigger because they can be connected to automatic disruption detection through weather data or a related API, but the solution remains open to all other disruptions.

For presentations, Tudlo should be framed clearly as an **educational continuity solution**, not a logistics tool. The core value is not disaster tracking by itself; it is recovery of learning continuity after interruption.

V. DEFINE SCOPE

A. Must-Have Features (MVP)

Feature	Description
Warm Start Onboarding	Teacher sets grade, subjects, and approximate curriculum position in under 5 minutes

Daily Progress Tracker	30–60 second daily lesson completion marking per subject
Line Graph of class closure	Graph to shows how each region have been impact with. Like it will based on how long each region or city has closure, so theres a downline trend So when i click a specific region on heatmap, it will show the line graph
Heat map	Heat map of region to city conditions of most impacted by disruptions like they have long closure, long closure means need more recovery, so the government can give resource to those affected like giving a personalized teacher that will actually go to that area.
PAGASA forecasting	Like theres a forecasting who will be affected if theres a typhoon will happen
CURRICULUM DEPED	The project will be basing on matatag curriculum so the app will align to the dates oon the project, so if the topic in 1st day is fraction and it was teached successfully, after that day there accumulating bar that will accumulated 1 bar for that days, so no need for teacher to manually click “done” after each teaching, one things she can is that when they encountered disruption is shen can click “disruption” thingy

Mark Disruption	One-tap curriculum position freeze with disruption type logging
Resume View	Post-disruption screen showing the exact lesson to resume per subject, with preserved teacher notes
Catch-Up Plan Generator	Rule-based calculation of a compressed schedule based on remaining school days and remaining lessons
Offline-First Operation	Full core functionality with zero internet connection
Filipino Language UI	Primary interface language in Filipino with English toggle
Teacher-Defined Lesson Granularity	Teacher manually defines how specific the “stopped point” should be, with finer granularity as a future refinement
Section-Level Recovery Plan	Recovery plan is generated for the class or section as a whole in the MVP

B. Nice-to-Have Features

Feature	Description
PAGASA Auto-Detection	Automatic disruption trigger when weather data indicates a typhoon affecting the teacher’s area
Recovery Rate Tracker	Post-resumption monitoring showing whether the class is on track to recover
At-Risk Subject Alerts	Notification when a subject falls behind the catch-up plan
Admin / Principal View	School-level summary of all teachers’ curriculum positions and disruption events
Disruption Heatmap (School Level)	Visual map showing disruption events and curriculum gap severity per school
Readiness Graph	A graph displayed when clicking a high-disruption heatmap area, showing readiness / recovery / learning continuity status over time
Recovery Threshold Indicator	A simple end-of-period check-in or confirmation button that indicates whether the class is recovering well
Student-Level Recovery View	Per-student recovery tracking for future refinement

Lesson-Granularity Controls	More detailed control over whether stopping points are tracked by lesson, subtopic, or competency
Numerical Disruption Analytics	Counts of disruptions, affected students, missed days, and regional comparisons used for thresholding and reporting

C. Future Features

Feature	Description
National Disruption Heatmap	DepEd-level dashboard showing real-time disruption density across regions
Mission Teacher Deployment Module	Tool to identify overwhelmed schools and coordinate volunteer teacher deployment
ML-Based Catch-Up Prioritization	Model that suggests which lessons can be compressed safely
DepEd LIS / BEIS Integration Sync	Sync Tudlo data with official DepEd systems
Student-Facing View	Student dashboard showing what to review at home
Multi-Language Support	Bicolano, Waray, and other regional language options

Curriculum Graph Engine	Prerequisite-aware lesson sequencing that prevents skipping foundational content
Dynamic Roadmap Planner	Long-term roadmap that spaces lessons across the remaining K–12 curriculum calendar and automatically adjusts after each disruption
Integrated LMS Add-on Mode	Optional layer that works on top of existing LMS platforms rather than replacing them

D. Output — MVP Scope Summary

Priority	Feature
MUST HAVE	Warm Start Onboarding
MUST HAVE	Daily Progress Tracker
MUST HAVE	Mark Disruption
MUST HAVE	Resume View
MUST HAVE	Catch-Up Plan Generator
MUST HAVE	Offline-First Operation

MUST HAVE	Filipino Language UI
MUST HAVE	Teacher-Defined Lesson Granularity
MUST HAVE	Section-Level Recovery Plan
NICE TO HAVE	PAGASA Auto-Detection
NICE TO HAVE	Recovery Rate Tracker
NICE TO HAVE	At-Risk Subject Alerts
NICE TO HAVE	Admin / Principal View
NICE TO HAVE	School-Level Disruption Heatmap
NICE TO HAVE	Readiness Graph
NICE TO HAVE	Recovery Threshold Indicator
NICE TO HAVE	Numerical Disruption Analytics
FUTURE	National Heatmap

FUTURE	Mission Teacher Deployment Module
FUTURE	ML Catch-Up Prioritization
FUTURE	DepEd LIS / BEIS Integration
FUTURE	Student-Facing View
FUTURE	Dynamic Roadmap Planner
FUTURE	Integrated LMS Add-on Mode

VI. DEFINE USERS

A. User Personas

Persona 1 — Primary User

Teacher Marivic Santos

Attribute	Detail
Age	38
Role	Grade 4 Public School Teacher

Location	Virac, Catanduanes, Region V
Device	Samsung Galaxy A14, Android 12
Connectivity	Intermittent LTE; unreliable during storms
Subjects Taught	Filipino, English, Math, Science, AP, MAPEH
Tech Comfort	Moderate; uses Messenger daily
Current Pain	After every typhoon, spends time asking students and colleagues where they left off, often restarting the unit to be safe
Motivation	Wants to help students catch up but feels overwhelmed after disruptions
Quote	“Pagkatapos ng bagyo, parang nagsisimula ulit tayo sa wala.”

Persona 2 — Secondary User

Principal Roberto Aquino

Attribute	Detail

Age	52
Role	School Principal
Location	Palo, Leyte, Region VIII
Device	Tablet + personal laptop
Connectivity	Stable office internet, often down post-typhoon
Responsibility	18 teachers, 640 students across Grades 1–6
Current Pain	No visibility into which teachers are behind after disruptions
Motivation	Wants accurate reports and support planning
Quote	“Alam ko na may problema, pero hindi ko alam kung gaano kalala.”

Persona 3 — Tertiary User

DepEd Division Superintendent, Dr. Ana Reyes

Attribute	Detail
Age	48
Role	Schools Division Superintendent
Location	Eastern Samar, Region VIII
Device	Laptop, desktop at division office
Connectivity	Stable office internet
Responsibility	Oversees 120+ schools across the division
Current Pain	Receives disaster reports but has no instructional continuity data
Motivation	Wants to deploy support to the right schools at the right time
Quote	“Sinasabi nila na okay na ang klase, pero hindi natin alam kung talagang okay.”

B. User Journey

Primary User Journey — Teacher Marivic

ENTRY POINT

Receives Tudlo link via school Messenger GC

▼

Installs as PWA from browser — no app store needed

▼

Completes Warm Start — 5 minutes

▼

Sets grade, subjects, and approximate lesson position

▼

NORMAL SCHOOL DAYS

End of each school day:
Opens Tudlo → taps lesson completion per subject (30–60 seconds total)

▼

TYPHOON ANNOUNCED

Taps “Mark Disruption” → selects Typhoon
Curriculum position frozen instantly

▼

SCHOOL CLOSED

Tudlo asks nothing further of Teacher Marivic

▼

SCHOOL REOPENS — Monday morning, 7:15 AM

Opens Tudlo → taps “Resume”
Sees exact lesson per subject + preserved notes
Generates catch-up plan in one tap

▼

Walks into class at 8:00 AM with a plan

END GOAL ACHIEVED

Secondary User Journey — Principal Roberto

Receives Tudlo admin access from Division Office

▼

Views school-level dashboard: all teachers' curriculum positions at a glance

▼

Post-typhoon, sees which teachers have marked disruption and which subjects are furthest behind

▼

Makes informed decisions on which teachers need support and which grades are most at risk

▼

Reports accurate data to Division Office

END GOAL ACHIEVED

Tertiary User Journey — Superintendent Dr. Ana Reyes

Accesses DepEd Division Dashboard via web browser

▼

Views regional disruption heatmap with school pins and severity cues

▼

Post-major-typhoon, identifies schools in critical gap territory

▼

Coordinates mission teacher deployment to those schools

▼

Tracks recovery rate week-over-week

▼

Submits evidence-based report to DepEd Regional Office

END GOAL ACHIEVED

VII. DEFINE THE BUSINESS MODEL

A. Target Audience Model

Primary Model: G2C (Government to Citizen/Civil Servant)
DepEd adopts and distributes Tudlo to public school teachers through official channels rather than individual teacher purchases.

Secondary Model: B2G (Business to Government)
The team or a future social enterprise pitches Tudlo as an institutional tool for government licensing or adoption partnership.

B. Value Proposition

User	Value Delivered
Teacher	Eliminates post-disruption curriculum guesswork and reduces the time spent rediscovering where the class stopped
Principal	Provides real-time visibility into curriculum position across all teachers without manual check-ins
DepEd Division / Regional	Produces aggregated recovery visibility for targeted intervention planning
Philippine Students	Preserves instructional continuity and reduces cumulative learning gaps

Why Better Than Existing Solutions:
Tudlo does not compete with DepEd Commons or Google Classroom. It fills a gap they do not address: knowing exactly where a class is in the curriculum at any moment, surviving disruptions, and recovering with precision.

C. Revenue Model

Given the public education context, traditional revenue is secondary to institutional adoption.

Model	Application
Government Licensing (Primary)	DepEd licenses Tudlo for deployment across public schools
NGO / Development Partner Funding	UNICEF Philippines, Save the Children, World Bank education programs support pilots
Freemium (Future)	Basic teacher features remain free; institutional dashboards may be advanced features

D. Cost Structure

Cost Item	Notes
Cloud Hosting	Free tier sufficient for hackathon MVP; scales later
Curriculum Data Preparation	One-time conversion of DepEd K–12 curriculum references into structured data

Weather API / Disruption Trigger	Public data source preferred for typhoon detection
Offline PWA Storage	Client-side local storage with sync when available
Mission Teacher Coordination (Future)	Operational cost absorbed by DepEd, not Tudlo

E. Key Partners

Partner	Role
DepEd	Primary institutional adopter and distribution channel
PAGASA or Weather Data Provider	Source for typhoon-triggered disruption detection
UNICEF Philippines	Potential pilot funding partner
Teach for the Philippines	Potential mission teacher deployment partner
SEAMEO INNOTECH	Research and validation partner

State Universities (UP, PNU)	Research partnership for impact measurement

F. Go-to-Market Strategy

Phase 1 — Hackathon (Now)
Present Tudlo as a clear educational continuity solution with an offline-first MVP and a typhoon-triggered demo path.

Phase 2 — Pilot (3–6 months post-hackathon)
Partner with 2–3 schools in a disruption-prone area through a university or NGO contact. Run one full quarter of data collection.

Phase 3 — Division Adoption (6–12 months)
Present pilot results to a DepEd division office and seek broader distribution through existing teacher communication channels.

Phase 4 — DepEd Central Engagement (12–24 months)
Submit pilot data to central offices and position Tudlo as a complementary tool to existing systems, not a replacement.

G. Scalability

Stage	Scope
Pilot	3–5 schools, 1 division, 1 region
Regional	All schools in the most disruption-prone regions
National	Typhoon-prone regions across the Philippines

Future	Similar disruption-prone education systems in other countries

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VIII. WRITE PRODUCT REQUIREMENTS (PRD)

A. Vision

To build the foundational curriculum continuity infrastructure for Philippine public education — ensuring that no typhoon, earthquake, flood, fire, evacuation, or other disruption permanently erases a teacher's knowledge of where a class stands, and that no school is forced to recover alone.

B. Goals

Goal	Metric
Eliminate post-disruption curriculum guesswork	Teacher resumes correct lesson within 60 seconds of opening Tudlo
Reduce wasted re-teaching time	Recover at least 3 instructional days per disruption event per classroom
Provide visibility into disruption recovery	Recovery graph or readiness indicator updates after a disruption
Enable school support planning	Principals can identify at-risk classes and teachers

Build the first empirical disruption-curriculum dataset	Collect usable recovery and disruption history over a school year

C. Features

Core Features (MVP)

F01 — Warm Start Onboarding
Teacher sets up their profile: grade level, subjects taught, and approximate current lesson position per subject. Tudlo maps this to the official DepEd curriculum reference and confirms the starting point.

F02 — Daily Progress Tracker
At the end of each school day, the teacher opens Tudlo and marks each subject as: Lesson Completed / Partially Covered / Not Reached Today. For partial coverage, the teacher defines where the class stopped at a teacher-selected granularity.

F03 — Mark Disruption
One-tap button on the home dashboard. Teacher selects disruption type (Typhoon / Flooding / Earthquake / Fire / Evacuation / Other). Tudlo timestamps the event and freezes the current curriculum snapshot for all subjects.

F04 — Resume View
When the teacher taps “School Has Resumed,” Tudlo displays a per-subject resumption card showing: exact lesson to start, last sub-topic covered, preserved notes, and number of lessons missed.

F05 — Catch-Up Plan Generator
Based on lessons missed, remaining school days in the term, and the current curriculum pace, Tudlo generates catch-up options and allows the teacher to confirm one. The plan must also account for **length of downtime**: if the class has been away from the lesson for too long, the system should recommend a brief refresher of earlier topics before moving forward.

F06 — Offline-First Operation
All core features function without internet connectivity. Data is stored locally and synced when connectivity returns.

F07 — Filipino Language UI
Primary interface text rendered in Filipino, with English toggle available.

F08 — Teacher-Defined Lesson Granularity
The teacher chooses how detailed the stopping point should be. Finer granularity remains a future refinement rather than a hard MVP requirement.

F09 — Section-Level Recovery Plan
The MVP recovery plan is generated per class section as a whole.

Secondary Features (Nice-to-Have)

F10 — PAGASA Auto-Detection
Background sync checks weather data and prompts the teacher to mark a disruption when a typhoon may affect the registered area.

F11 — Recovery Rate Tracker
Tracks recovery against the confirmed catch-up plan.

F12 — At-Risk Subject Alerts
Surfaces subjects that are falling behind recovery pace.

F13 — Principal Dashboard
Read-only school-level web view for principals.

F14 — School-Level Disruption Heatmap
Visual map showing which classes are disrupted and how severe the gap is.

F15 — Readiness Graph
Clicking a high-disruption heatmap area opens a recovery/readiness graph, similar in concept to the attached sample chart, showing how the class is recovering over time.

F16 — Recovery Threshold Indicator
A simple end-of-period check-in or confirmation button that indicates whether the class is recovering well. This serves as the practical “high point” or threshold signal.

F17 — Numerical Disruption Analytics
Counts of disruptions, missed days, and affected classes are used as supporting inputs for recovery thresholds and reporting. These are secondary metrics, not the core MVP focus.

Future Features (Roadmap)

F18 — National Disruption Heatmap
DepEd-level dashboard aggregating all school-level disruption data.

F19 — Mission Teacher Deployment Module
Identifies overwhelmed schools and tracks support assignments.

F20 — Dynamic Roadmap Planner
A long-term planning layer that spaces lessons across the curriculum calendar using K–12 references and automatically adjusts after each disruption.

F21 — DepEd System Integration
API connections to LIS and BEIS.

F22 — Student-Facing View
Student dashboard showing review tasks and catch-up status.

F23 — Curriculum Graph Engine
Prerequisite-aware sequencing to prevent skipping foundational content.

F24 — Integrated LMS Add-on Mode
Optional mode that works with existing LMS platforms if integration becomes the preferred adoption path.

D. Functional Requirements

ID	Requirement	Priority
FR01	System shall allow teacher to complete onboarding in under 5 minutes	Must Have
FR02	System shall store curriculum position locally on device without internet	Must Have
FR03	System shall allow teacher to mark daily lesson progress in under 60 seconds	Must Have
FR04	System shall freeze curriculum position	Must Have

	snapshot on disruption marking	
FR05	System shall display exact resume lesson per subject within 3 taps of opening post-disruption	Must Have
FR06	System shall generate at least 2 catch-up plan options based on remaining school days and downtime length	Must Have
FR07	System shall sync all local data to cloud when internet connectivity is restored	Must Have
FR08	System shall support Filipino language as primary UI language	Must Have
FR09	System shall allow teacher to manually define lesson stopping point granularity	Must Have
FR10	System shall support disruption types beyond typhoon	Must Have
FR11	System shall query weather or disruption data to	Nice to Have

	suggest typhoon-related marking when available	
FR12	System shall track daily recovery rate against confirmed catch-up plan	Nice to Have
FR13	System shall alert teacher when subject recovery falls behind plan for 3+ days	Nice to Have
FR14	System shall provide principal with read-only school-level curriculum dashboard	Nice to Have
FR15	System shall aggregate disruption events into a school-level heatmap for admin access	Nice to Have
FR16	System shall display readiness graph when a high-disruption area is selected	Nice to Have
FR17	System shall provide a recovery threshold indicator at the end of a recovery period	Nice to Have

FR18	System shall calculate how many lessons or modules remain to be caught up	Nice to Have
FR19	System shall support future integration with DepEd systems	Future
FR20	System shall support future student-facing recovery views	Future

E. Non-Functional Requirements

ID	Requirement	Detail
NFR01	Offline Capability	100% of core features (FR01–FR10) must function with zero internet connection
NFR02	Load Time	App must launch and reach home screen in under 3 seconds on a low-end Android device
NFR03	Storage Footprint	Total app storage including curriculum data must not exceed 50MB on device

NFR04	Compatibility	Must function on Android 9 and above, with Chrome browser support
NFR05	Data Persistence	Local data must survive app closure, device restart, and a 30-day offline period
NFR06	Sync Reliability	On connectivity restoration, queued local data must sync without loss or duplication
NFR07	Usability	Any primary task must be completable in 3 taps or fewer from the home screen
NFR08	Language	All primary UI text in Filipino; no untranslated English strings in core flows
NFR09	Security	Teacher data isolated by account; no cross-school access without admin role
NFR10	Scalability	Backend architecture must support scaling from 10 to 10,000 concurrent teachers without redesign
NFR11	Availability	Cloud sync layer targets 99% uptime; local layer is

		always available regardless of cloud status
NFR12	Accessibility	Minimum font size 16px; icon + text labels on all primary actions; sufficient color contrast

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IX. VALIDATION AND EVIDENCE PLAN

A. Validation Strategy

The project should be validated through:

- Teacher interviews
- Principal interviews
- Anonymous testimonials
- DepEd reports
- News reports about school disruptions
- Supporting research on disaster-related learning loss and recovery

B. What Counts as Evidence of Success

Tudlo is considered effective if it can show:

- Teachers resume the correct lesson more quickly after a disruption
- Teachers spend less time reconstructing curriculum position
- Recovery plans become clearer and more consistent
- School-level leaders can see where support is needed
- The readiness graph or recovery indicator shows improvement after assistance is provided

C. Interpretation of Recovery

Recovery should be treated as the point where a class has returned to a manageable instructional path after disruption, as shown by the recovery indicator, readiness graph, and teacher confirmation.

X. SUMMARY OF MVP POSITIONING

Tudlo is an offline-first curriculum continuity tool for Philippine public schools. Its catch-up logic is downtime-aware, so prolonged interruptions can trigger a short refresher of previous topics before the class resumes new lessons. Its catch-up logic is downtime-aware, so prolonged interruptions can trigger a short refresher of previous topics before the class resumes new lessons.

It is primarily a teacher workflow tool.

Typhoons are the primary MVP case because they can be supported by automated disruption detection, but the product must remain open to all other disruptions.

The MVP focus is:

- manual teacher tracking of where lessons stopped
- immediate resume instructions
- class-level recovery planning
- optional typhoon-based auto prompt
- readiness and recovery visualization for repeated disruptions

The roadmap extends into school-level and DepEd-level recovery intelligence, but those are future phases, not MVP requirements.